

$u(t_n, x^j(t_n))$ the exact solution computed at the exact departure point x_{e_j} of the trajectory Γ_j^{n+1} ; and $u_{d_j}^n := u(t_n, x_{d_j})$ the exact solution computed at the approximated departure point x_{d_j} of the trajectory Γ_j^{n+1} . The linear operator defined along Γ_j^{n+1} is denoted by $L_{x^j(t)}$. Some of these definitions are presented in Figure 1.

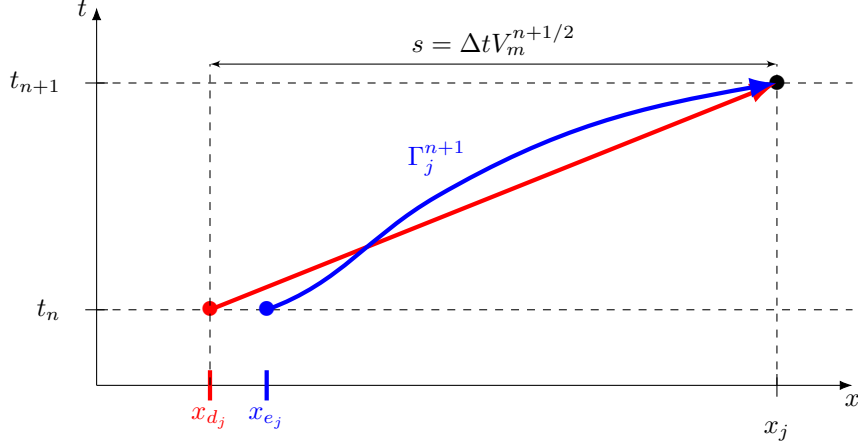


FIGURE 1. Exact Lagrangian trajectory Γ_j^{n+1} (in blue) in $[t_n, t_{n+1}]$, and estimated trajectory provided by SETTLS (in red), where $V_m^{n+1/2}$ is an estimation of the velocity field on its midpoint $(t_{n+1/2}, x_m)$ (see Eq. (3.10)).

This discretization leads to a system of uncoupled ODEs which are approximated by (4.1) as

$$U_j^{n+1} = \varphi_0(\Delta t L_j) U_{*j}^n, \quad j = 0, \dots, P \quad (4.4)$$

where w_{*j} , for a given function w defined on the spatial grid, denotes interpolation to the estimated departure point x_{d_j} , and $L_j := L_{x^j(t_{n+1})} = L(x_j)$.

Using cubic Lagrange interpolation to the approximated departure point x_d , it is well known that $w_{*j} = w_{d_j} + \mathcal{O}(\Delta x^4)$ if w is smooth enough. It is also well known that SETTLS (3.10) provides a second-order approximation to the solution computed at the true departure point x_{e_j} , more precisely, $u_{d_j} = u_{e_j} + \mathcal{O}(\Delta t^3) + \mathcal{O}(\Delta t \Delta x^2)$ [10, Chapter 7]. Combining these results, we get, in the case of SETTLS using cubic Lagrange interpolation,

$$u_{*j} = u_{d_j} + \mathcal{O}(\Delta x^4) = u_{e_j} + \mathcal{O}(\Delta t^3) + \mathcal{O}(\Delta t \Delta x^2) + \mathcal{O}(\Delta x^4) \quad (4.5)$$

From (4.4), it is clear that the exponential function is computed at the arrival point $x_j = x^j(t_{n+1})$ of each Lagrangian trajectory. In other words, the integral in $P_n(t_{n+1})$ (see Eq. (3.14)) is approximated by considering the integrand to be constant in $[t_n, t_{n+1}]$ and equal to $L_{x^j(t_{n+1})}$. Note that it is coherent with the interpretation of the order in which the operators are applied in (4.1): the solution is obtained at t_{n+1} through interpolation, then the linear operator, also evaluated at t_{n+1} , is applied to it. This statement also allows us to evaluate the error in the approximation of the integral in (3.14). From the theory of numerical integration using Newton-Cotes formulae (see e.g., [38, chapter 3]), it is well known that

$$\int_{t_n}^{t_{n+1}} L_{x^j}(s) ds = \Delta t L_{x^j}(t_{n+1}) + C_{1,j} \Delta t^2 = \Delta t L_j + C_{1,j} \Delta t^2 \quad (4.6)$$

for a given constant $C_{1,j}$ depending on the first derivative of L along the trajectory Γ_j^{n+1} . Replacing (4.6) in (3.14) and denoting by $P_{n_j}(t)$ the integration factor (3.14) defined along the Lagrangian trajectory Γ_j^{n+1} , we obtain

$$P_{n_j}(t_{n+1}) = e^{-\Delta t L_j} e^{-C_{1,j} \Delta t^2}. \quad (4.7)$$

We are now able to obtain the local truncation error τ_j^{n+1} at (t_{n+1}, x_j) of (4.1) applied to (4.2). The exact solution of (4.3) along Γ_j^{n+1} reads

$$u_j^{n+1} = u(t_{n+1}, x^j(t_{n+1})) = P_{n_j}^{-1}(t_{n+1}) u(t_n, x^j(t_n)) = P_{n_j}^{-1}(t_{n+1}) u_{e_j}^n. \quad (4.8)$$

Replacing the exact solution into the j -th equation of (4.4), and using (4.5), (4.7) and (4.8), yields

$$\begin{aligned} \tau_j^{n+1} &= \frac{1}{\Delta t} \left[u_j^{n+1} - \varphi_0(\Delta t L_j) u_{*j}^n \right] \\ &= \frac{1}{\Delta t} \left[P_{n_j}^{-1}(t_{n+1}) u_{e_j}^n - e^{\Delta t L_j} u_{*j}^n \right] \\ &= \frac{1}{\Delta t} \left[e^{\Delta t L_j} e^{C_{1,j} \Delta t^2} u_{e_j}^n - e^{\Delta t L_j} \left(u_{e_j}^n + \mathcal{O}(\Delta t^3) + \mathcal{O}(\Delta t \Delta x^2) + \mathcal{O}(\Delta x^4) \right) \right] \\ &= \frac{1}{\Delta t} \left[e^{\Delta t L_j} (1 + C_{1,j} \Delta t^2 + \mathcal{O}(\Delta t^4)) u_{e_j}^n - e^{\Delta t L_j} \left(u_{e_j}^n + \mathcal{O}(\Delta t^3) + \mathcal{O}(\Delta t \Delta x^2) + \mathcal{O}(\Delta x^4) \right) \right] \\ &= \mathcal{O}(\Delta t) + \mathcal{O}(\Delta x^2) + \mathcal{O}\left(\frac{\Delta x^4}{\Delta t}\right), \end{aligned} \quad (4.9)$$

confirming that the scheme (4.1) provides a first-order approximation to (4.2). Note that this first-order is due exclusively to the term $e^{C_{1,j} \Delta t^2}$, which arises from the approximation to the integral in the integration factor $P_{n_j}(t_{n+1})$; all other approximations are second-order accurate.

5. SECOND-ORDER SEMI-LAGRANGIAN EXPONENTIAL INTEGRATION METHOD

The truncation error analysis presented above indicates that the approximation of the integral in the integration factor (3.14) is responsible for the first-order accuracy of the SE12 scheme. Therefore, we provide an alternative formulation for this approximation which is effectively a second-order one. A natural approach is to consider an $\mathcal{O}(\Delta t^3)$ approximation the integral; for instance, the trapezoidal rule. In the scalar case, using the notation defined in the previous section, it reads

$$\int_{t_n}^{t_{n+1}} L_{x^j}(s) ds = \frac{\Delta t}{2} [L_{x^j}(t_{n+1}) + L_{x^j}(t_n)] + C_{2,j} \Delta t^3 = \frac{\Delta t}{2} [L_j + L_{e_j}] + C_{2,j} \Delta t^3 \quad (5.1)$$

for a constant $C_{2,j}$ depending on the second derivative of L along Γ_j^{n+1} , and $L_{e_j} := L_{x^j}(t_n) = L(x_{e_j})$. Then, the integration factor (3.14) along the Lagrangian trajectory Γ_j^{n+1} satisfies

$$P_{n_j}(t_{n+1}) = e^{-\int_{t_n}^{t_{n+1}} L_{x_j(s)} ds} = e^{-\frac{\Delta t}{2}[L_j + L_{e_j}] - C_{2,j}\Delta t^3} = e^{-\frac{\Delta t L_j}{2}} e^{-\frac{\Delta t L_{e_j}}{2}} e^{-C_{2,j}\Delta t^3}, \quad (5.2)$$

yielding the scheme

$$U^{n+1} = \varphi_0\left(\frac{\Delta t L}{2}\right) \left[\varphi_0\left(\frac{\Delta t L}{2}\right) U^n \right]_*^n \quad (5.3)$$

Note that the last equality in (5.2) holds because L_j and L_{e_j} are scalars. The order in which the operators are applied in (5.3) (application of the exponential inside the brackets, followed by a spatial interpolation, followed by a second application of the exponential) arises from (5.2) combined with a similar interpretation as in the first-order scheme: half time step with the linear operator, evaluated at t_n (*i.e.*, at the departure point, L_{e_j}), is applied to the solution computed at t_n ; then the solution is obtained at t_{n+1} through interpolation; finally, half time step with the linear operator at t_{n+1} (*i.e.*, at the arrival point, L_j) is applied to the solution.

In the vectorial case

$$\frac{DU}{Dt} = LU \quad (5.4)$$

with $U \in \mathbb{R}^N$, $L \in \mathbb{R}^{N \times N}$, we propose an analogous scheme, constructed directly from (5.3):

$$U^{n+1} = \varphi_0\left(\frac{\Delta t L}{2}\right) \left[\varphi_0\left(\frac{\Delta t L}{2}\right) U^n \right]_*^n \quad (5.5)$$

Note, however, that an analogous derivation is not necessarily valid. Indeed, we have, in general,

$$e^{-\frac{\Delta t}{2}[L_j + L_{e_j}]} \neq e^{-\frac{\Delta t}{2}L_j} e^{-\frac{\Delta t}{2}L_{e_j}} \quad (5.6)$$

except if L_j and L_{e_j} commute. In Section 5.1 we demonstrate the second-order accuracy of (5.3), in the scalar case, and of (5.5) in the vectorial case when L_j and L_{e_j} commute; the numerical simulations presented in Section 7 indicate that the second-order accuracy is preserved even if this commutation hypothesis does not hold.

Applying (5.5) to the full PDE (3.1), we obtain the methods:

$$U_{\text{SE21}}^{n+1} = \varphi_0\left(\frac{\Delta t L}{2}\right) \left[\varphi_0\left(\frac{\Delta t L}{2}\right) U^n \right]_*^n + \varphi_0(\Delta t L) \left[\Delta t \psi_1(\Delta t L) \tilde{N}(U^n) \right]_*^n, \quad (5.7a)$$

$$U_{\text{SE22}}^{n+1} = U_{\text{SE21}}^{n+1} + \Delta t \varphi_0(\Delta t L) \left[\psi_2(\Delta t L) \tilde{N}(U_{\text{SE21}}^{n+1}) - \left(\psi_2(\Delta t L) \tilde{N}(U^n) \right)_*^n \right] \quad (5.7b)$$

where the Lagrangian trajectories are estimated as in the SE11 and SE12 schemes, using SETTLS (Eq. (3.10)).

As proved in Section 5.1 and/or verified through numerical simulations in Section 7, (5.7b) provides a second-order approximation to (3.1). Therefore, the names of the methods were chosen following the same reasoning as before. In (5.7a), a second- and a first-order discretizations of the linear and nonlinear terms, respectively, resulting in a global first-order approximation; in (5.7b), a global second-order accuracy is ensured by second-order discretizations of both linear and nonlinear terms.

Note that the proposed discretization, using the approximation (5.2) to the integration factor, is applied only to the linear term, *i.e.*, the nonlinear one is discretized as in SE11 (eq. (3.15a))